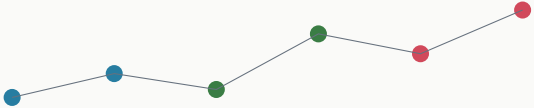
Representation Geometry in Deep Learning

Static geometry, abstraction, and dynamics



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One behavioral result, many internal solutions

Network A

- same held-out accuracy
- hidden units have no obvious labels



Network B

- same held-out accuracy
- a different hidden basis



What is represented, and in what form?

Learning goals

By the end of the session, we will be able to:

- summarize a representation with a dissimilarity matrix (RSA);
- distinguish decodability from cross-condition generalization;
- characterize abstraction with CCGP and parallel coding directions;
- relate abstraction to shattering dimensionality;
- trace DSA from delay coordinates to aligned evolution operators;
- distinguish RSA, CCGP, and DSA.

Three questions, three method families

RSA

- Which conditions are similar or dissimilar?
- Do two models share the same static geometry?

CCGP

- Does one linear readout transfer to unseen conditions?
- Which geometric structure supports that transfer?

DSA

- Do state transitions of the hidden states follow similar rules?
- Are the dynamics equivalent up to coordinate alignment?

Notation at a glance

Symbol	Meaning
$\mathbf{z}_n$	representation of sample n
$\boldsymbol{\mu}_c^{(\ell)}$	mean representation for condition c at layer ℓ
$D^{(\ell)}$	RDM at layer ℓ
CCGP	cross-condition generalization performance
PS	parallelism score of coding directions
SD	shattering dimensionality
H_x, H_y	delay-coordinate data matrices
A, B	fitted linear evolution operators
d_{DSA}	dynamical similarity distance

A representation is a set of response vectors

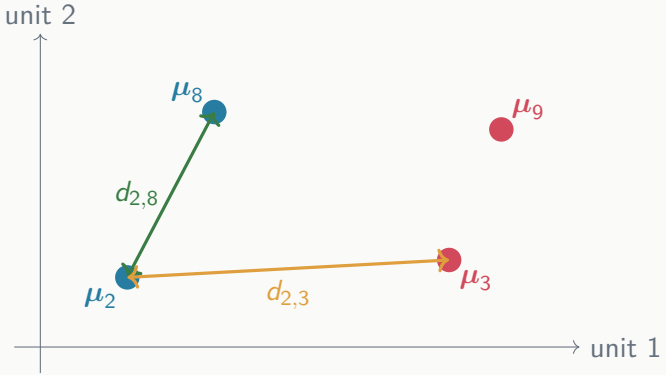
For sample n with input $\mathbf{x}_n$ and condition label y_n, at layer ℓ:

$$\mathbf{z}_n^{(\ell)} = f^{(\ell)}(\mathbf{x}_n) \in \mathbb{R}^{p_\ell}.$$

For condition c with N_c samples, summarize repeated samples by their mean:

$$\boldsymbol{\mu}_c^{(\ell)} = \frac{1}{N_c} \sum_{y_n = c} \mathbf{z}_n^{(\ell)}.$$

Coordinates describe a map; distances describe its geometry



- Geometry can remain stable when the coordinate basis changes.
- Pairwise distances provide a coordinate-free summary for Euclidean transformations $O(N)$.

Running example: three labels for each digit

Digit	Identity	Parity	Magnitude
2	two	even	small
3	three	odd	small
8	eight	even	large
9	nine	odd	large

- Small: digits 0–4; large: digits 5–9.
- Parity and magnitude are crossed variables, meaning we have four conditions.

Step 2: compute a dissimilarity

At a fixed layer ℓ, compare condition means:

$$d_{ij}^{(\ell)} = \|\boldsymbol{\mu}_i^{(\ell)} - \boldsymbol{\mu}_j^{(\ell)}\|_2 \quad \text{or} \quad d_{ij}^{(\ell)} = 1 - \mathrm{corr}(\boldsymbol{\mu}_i^{(\ell)}, \boldsymbol{\mu}_j^{(\ell)}).$$

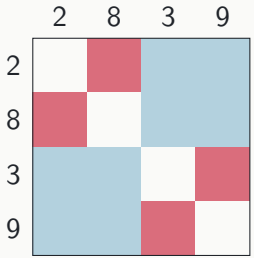
- Euclidean distance retains response magnitude.
- Correlation distance emphasizes response profile after centering and scaling.
- Each layer yields its own set of pairwise distances.

Step 3: arrange distances in an RDM

$$D^{(\ell)} = \begin{bmatrix} 0 & d_{12}^{(\ell)} & \cdots & d_{1K}^{(\ell)} \\ d_{21}^{(\ell)} & 0 & \cdots & d_{2K}^{(\ell)} \\ \vdots & \vdots & \ddots & \vdots \\ d_{K1}^{(\ell)} & d_{K2}^{(\ell)} & \cdots & 0 \end{bmatrix}$$

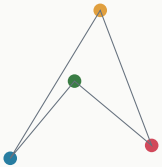
- One RDM per layer ℓ
- Symmetric for symmetric distances
- Zero diagonal for ordinary distances
- $K(K-1)/2$ unique entries

An RDM makes grouping visible



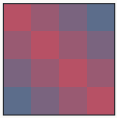
- For a network that discriminates for parity,
- Red: smaller dissimilarity
- Blue: larger dissimilarity

RSA compares relations, not units



Network layer





Common comparison space

Different feature counts and bases can still yield comparable RDMs. This allows RSA to compare artificial neural networks with brain regions.

Abstraction means transfer across task conditions

A binary variable is represented abstractly when a linear rule learned from some conditions generalizes to new conditions on both sides of that variable.

Target variable

- parity: even versus odd
- the label the decoder must predict

Held-out context

- magnitude: small versus large
- the source of new conditions

A dichotomy states the geometric hypothesis

Let $\mathcal{C} = \{1, \ldots, K\}$ index condition means. A balanced dichotomy divides them into two equally sized groups:

$$\mathcal{C} = \mathcal{C}_+ \cup \mathcal{C}_-, \qquad |\mathcal{C}_+| = |\mathcal{C}_-| = K/2.$$

- Each dichotomy assigns a binary label to every condition.
- Parity, magnitude, and other balanced labelings are different dichotomies.
- With eight conditions, there are $\frac{1}{2}\binom{8}{4} = 35$ balanced dichotomies.

The analysis asks which labelings are decodable, and which are supported by a transferable geometry.

Balanced-dichotomy analysis: [3], "Measuring Abstraction."

CCGP withholds conditions, not merely trials

For one dichotomy $(\mathcal{C}_+, \mathcal{C}_-)$:

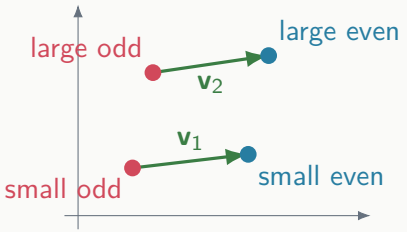
1. choose training conditions from both sides;
2. fit a linear classifier using only those conditions;
3. evaluate on conditions that the classifier never encountered;
4. repeat across admissible train-test assignments and average.

$$\mathrm{CCGP} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \mathrm{Acc}\left(h_s;\, \mathcal{C} \setminus \mathcal{C}_s^{\mathrm{train}}\right).$$

For parity across magnitude, the two assignments are small $\rightarrow$ large and large $\rightarrow$ small.

CCGP procedure: [3], Fig. S1A and STAR Methods.

Parallel coding directions make a rule reusable



For paired conditions,

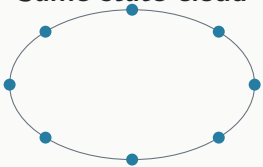
$$\mathbf{v}_j = \frac{\boldsymbol{\mu}_{+,j} - \boldsymbol{\mu}_{-,j}}{\|\boldsymbol{\mu}_{+,j} - \boldsymbol{\mu}_{-,j}\|_2}.$$

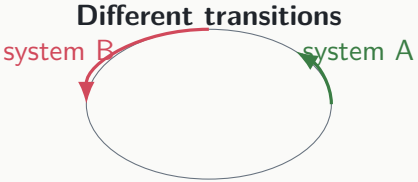
The parallelism score (PS) averages pairwise cosines and selects the best one-to-one pairing.

High PS describes aligned condition means. CCGP also depends on spacing and trial-level noise.

Parallelism score: [3], Fig. S1B–C and STAR Methods.

Static geometry can omit the transition rule





Pairwise state geometry does not specify how one state leads to the next.

DSA compares evolution laws after alignment

Consider two systems observed along matched conditions and trials:

$$\dot{\mathbf{x}} = f(\mathbf{x}, t), \qquad \dot{\mathbf{y}} = g(\mathbf{y}, t).$$

DSA asks whether their learned evolution operators agree after an admissible change of coordinates.

- Input: uniformly sampled trajectories, organized by condition and trial
- Common space: delay coordinates reduced to the same rank r
- Output: an operator distance; lower values indicate more similar dynamics

$$\text{trajectories} \longrightarrow \text{delay coordinates} \longrightarrow A_x, A_y \longrightarrow \text{alignment} \longrightarrow d_{\mathrm{DSA}}$$

Dynamical similarity analysis: [4], Secs. 2.1–2.2.

Delay coordinates recover state carried by history

With p stored samples separated by τ, each time point becomes

$$\mathbf{h}_t = \begin{bmatrix} \mathbf{x}_t^\top & \mathbf{x}_{t-\tau}^\top & \cdots & \mathbf{x}_{t-(p-1)\tau}^\top \end{bmatrix}^\top \in \mathbb{R}^{np}.$$

- Stacking the $\mathbf{h}_t$ produces a delay-coordinate matrix H_x.
- History can reveal latent state that is absent from one observation $\mathbf{x}_t$.
- Under Takens' assumptions, a sufficient number of delay observations reconstruct a state space diffeomorphic to the original system.
- Delays are formed within each uniformly sampled trial.

Delay reconstruction and HAVOK: [4], Sec. 2.1.

DSA aligns vector fields, not point clouds

Ordinary Procrustes

$$d_{\text{shape}}(X, Y) = \min_{C \in O(r)} \|X - CY\|_F.$$

A point cloud changes as $Y \mapsto CY$.

Procrustes over vector fields

$$d_{\text{DSA}}(A_x, A_y) = \min_{C \in O(r)} \|A_x - CA_yC^{-1}\|_F.$$

A linear evolution operator changes by conjugation.

- For orthogonal C, $C^{-1} = C^\top$.
- The paper also uses a normalized angular form of the operator distance.
- In the notebook's fully observed linear case, A_x and A_y are fitted directly, so delay embedding is omitted.

Vector-field alignment: [4], Eqs. 4–6.

Static and dynamic methods answer different questions

Method	Main question	Requires time?	Typical output
RSA	Which conditions have similar geometry?	no	RDM similarity
CCGP	Does a variable transfer across context?	no	transfer accuracy
PS / SD	Which directions align / remain readable?	no	geometric scores
DSA	Do systems follow similar dynamics?	yes	aligned operator distance

Choose the method from the claim, not from the data format alone.

Predictions to record before running code

1. Which digit representation will best match parity and magnitude model RDMs?
2. Which digit representation will produce the highest CCGP?
3. Why can rotating all states preserve DSA while changing coordinates?
4. Can two systems visit similar states but follow different transition rules?

Write one prediction and one reason.

Takeaways

- RSA compares pairwise geometry among specified conditions.
- CCGP measures whether a selected variable transfers to unseen conditions.
- DSA compares delay-embedded evolution operators after coordinate alignment.
- A DSA score depends on the sampling, embedding, rank, and fitted-model quality.
- Similar static geometry does not determine similar dynamics.

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[3] S. Bernardi, M. K. Benna, M. Rigotti, J. Munuera, S. Fusi, and C. D. Salzman.
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